



# Circulating plasma microRNA expression profile: potential minimally invasive biomarker for early detection of coronary artery disease development

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## Abstract

**Background** The prevalence of coronary artery disease (CAD) is increasing among young adults. To improve CAD diagnosis, microRNAs are being explored as potential minimally invasive biomarkers. The aim of this study was to evaluate circulating microRNA (miRNA) expression profiles and assess their value in predicting the development of early-onset CAD.

**Methods and results** A total of 108 patients with early- and late-onset CAD and 29 individuals without CAD were included, and their miRNA expression was evaluated. The diagnostic value of differentially expressed miRNAs across the subgroups was tested by logistic regression models and ROC curve analysis. A total of 287 different circulating miRNAs were analysed following sequencing and preprocessing. Seven miRNAs (miR-10b-5p, miR-29c-3p, miR-142-5p, miR-320b, miR-451a, miR-486-3p, and miR-625-3p) were found to be differentially expressed across all the study groups, four of which (miR-142-5p, miR-29c-3p, miR-451a, and miR-486-3p) were significantly downregulated in the late-onset CAD group compared with the control group. ROC analysis demonstrated that the combination of the seven miRNAs had high diagnostic accuracy, with an AUC of 0.9924 for distinguishing late-onset CAD from the other groups, and moderate accuracy, with an AUC of 0.8235 for distinguishing early-onset CAD from the other groups.

**Conclusions** A combination of seven circulating miRNAs (miR-10b-5p, miR-29c-3p, miR-142-5p, miR-320b, miR-451a, miR-486-3p, and miR-625-3p) is a promising biomarker panel for CAD diagnosis, distinguishing between early-onset and late-onset disease. While the panel demonstrated high accuracy in classifying late-onset CAD, its ability to predict early-onset CAD requires further validation. Larger, independent populations are needed to validate the predictive ability of the panel for early disease detection, confirm these findings, and improve generalizability.

**Keywords** MicroRNA · Coronary artery disease · Atherosclerosis · Premature coronary artery disease

## Introduction

The prevalence of coronary artery disease (CAD) is increasing among young adults [1], and it may have a poor long-term prognosis [2, 3]. The aetiology of CAD is atherosclerosis driven by conventional lifestyle-related risk factors, such as smoking, obesity, diabetes mellitus, hypertension and dyslipidaemia. In recent decades, an increase in unhealthy lifestyle risk profiles has been observed among young adults, worsening their cardiometabolic health. The prevalence of obesity, which is associated with poor dietary habits, doubled among young adults in Finland between 2000 and 2017 [4]. A significant presence of dyslipidaemia [5] and hypertension [5, 6] has been reported among young

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patients in Europe and the United States (US). Compared with that in 2011, a dramatic increase in type 2 diabetes among young adults was documented in Malmö, Sweden, with a 119.6% increase observed in 2018 [7]. The majority of young CAD patients have a combination of several lifestyle-related risk factors [8]. However, a diagnostic challenge is young patients who develop CAD without the presence of traditional risk factors. Evidence suggests that non-traditional risk factors contribute to increased cardiovascular event risk in younger patients [9]. Therefore, in addition to the well-established traditional biomarkers of CAD, novel molecular and cellular markers for improved diagnostic and prognostic capabilities are being explored.

MiRNAs are small non-coding single-stranded RNA molecules involved in the post-transcriptional regulation of gene expression [10]. Specific miRNAs play crucial roles in the pathogenesis of atherosclerosis, influencing various cellular functions, lipid metabolism, and inflammatory responses. Kumar D et al. reported that the expression of miRNA-133b is downregulated and that of miRNA-21 is upregulated in the plasma samples of CAD patients and that the expression is correlated with disease severity [11]. A study by Wang et al. revealed that specific miRNAs may be dysregulated in early coronary atherosclerotic plaques [12]. Moreover, dysregulation of miRNA-146a\_1 and miRNA-342\_1 can predict major adverse cardiac events in early-onset CAD patients, whereas miRNA-145\_1 can be used to predict the need for revascularization [13]. Nonetheless, whether circulating miRNAs can serve as effective diagnostic biomarkers for predicting early-onset CAD remains uncertain.

The aim of this study was to evaluate the circulating miRNA expression profiles across CAD patient subgroups on the basis of disease onset and to explore the potential of these profiles in predicting early-onset CAD, both individually and through a combined miRNA model.

## Materials and methods

### Patients and blood sample collection

This prospective study, conducted at a single centre, received approval from the local ethics committee, and the study protocol was approved by an institutional review board. The study conforms to the ethical guidelines of the 1975 Declaration of Helsinki. Patients were enrolled after signed informed consent was obtained. All patients enrolled in this study were hospitalized for elective coronary angiography because their symptoms were indicative of stable coronary artery disease. Patients were divided into three groups according to their age and coronary angiography

findings: the control group ( $n=29$ ), early-onset CAD group ( $n=80$ ) and late-onset CAD group ( $n=28$ ). Patients were included in the control group if no atherosclerotic lesions were detected on coronary angiography and if their age was  $<55$  years for males and  $<65$  years for females. Patients with atherosclerotic lesions were divided into the early-onset CAD group if the patients' age was  $<55$  years for male patients and  $<65$  years for female patients. Conversely, if coronary angiography confirmed atherosclerotic lesions in male patients aged  $\geq 55$  years and female patients  $\geq 65$  years without any history of early-onset CAD (including anamnesis of premature myocardial infarction), these patients were assigned to the late-onset CAD group. Patients with cardiac or non-cardiac illness with a life expectancy of less than one year, acute coronary syndrome, congestive heart failure (NYHA III-IV), anamnesis of uncontrolled hypertension, obesity  $\geq$  grade 2, familial hypercholesterolemia, diabetes mellitus, glucose lowering medication use, and significant renal disease with a glomerular filtration rate  $<30$  ml/min/ $m^2$  were excluded from the study. Fasting blood samples were obtained from all the study patients to determine plasma glucose, liver and kidney markers, and lipid levels. Patients' baseline demographic and clinical characteristics were collected.

### Evaluation of MiRNA expression

Patients' fasting blood samples for the isolation of miRNA were obtained before or 24 h after heparin administration. Blood samples were collected in EDTA-containing venous blood collection tubes (Becton Dickinson, USA), and plasma separation was performed by rapid centrifugation for 20 min at 4 °C. The supernatant was then stored at  $-80$  °C in RNase/DNase-free tubes in 500  $\mu$ L aliquots. Frozen plasma samples were sent to CeGaT GmbH (Center for Genomics and Transcriptomics), a genetic diagnostics and sequencing service provider (Tübingen, Germany), where miRNA isolation, library preparation and sequencing were performed according to their internal procedures. Total RNA containing miRNA was purified according to the instructions of the miRNeasy Serum/Plasma Advanced kit protocol (Qiagen, Valencia, CA). Libraries were prepared using 5  $\mu$ L of isolated eluate with the QIAseq miRNA Library Kit (Qiagen, Valencia, CA). MiRNA analysis was performed using next-generation sequencing (NGS) because of its ability to provide comprehensive and unbiased detection of both known and novel miRNAs, which was essential for achieving our initial discovery objectives. Primary bioinformatics demultiplexing analysis was performed with Illumina bcl2fastq (version 2.20). Adapters were trimmed with Skewer (version 0.2.2) [14]. Quality trimming of the reads was not performed. The quality of the FASTQ files

was analysed with FastQC v0.12.1 (Andrews 2010) [15]. To remove adapter sequences and filter sequences on the basis of size, miRDeep2 v2.0.1.3 was used. Only reads consistent with the miRNA length of 18 nt were retained for further analysis. The final post-processing outcomes, including trimmed sequences and quality metrics, were aggregated and visualized using MultiQC v1.9. Thereafter, the sequencing reads were aligned to the human genome (GRCh38.p13, Ensembl) to allow for the detection of both known and potentially novel miRNAs. The mature and hairpin miRNA sequences used for identification were sourced from miR-Base (<http://www.mirbase.org/>, release 22.1, accessed on 1 February 2024). Mapping and miRNA identification were performed using Bowtie v1.0.0, a component of the miRDeep2 v0.1.3 suite. The quantification of miRNAs was performed using miRDeep2.pl, which integrates aligned reads with reference miRNA sequences to identify and quantify both known and novel miRNAs. Median read counts were calculated across multiple miRNA precursors to correct for redundancy caused by alternative precursor processing. To enhance the robustness of the differential expression analysis, an initial filtering step was applied, in which genes whose expression was consistently low were excluded; only miRNAs whose median expression was  $\geq 10$  in at least one group were retained for further analysis. ComBat-seq from the sva package (v3.50.0) was employed to adjust for batch effects between the sequencing batches. The plots were created using ggplot2 [16] in R (version 4.0.4) (R Core Team 2015).

## Statistical analysis

Data are presented as the mean  $\pm$  SD for normally distributed continuous variables and median (IQR) for nonnormally distributed variables, as determined by the Shapiro–Wilk test for normality. Categorical variables are expressed as counts (percentages). Comparisons between groups for continuous variables were performed using independent t tests for normally distributed variables and the Mann–Whitney U test for nonnormally distributed variables. For categorical variables, comparisons were conducted using the chi-square test. The R package edgeR (version 4.0.16) was used for Trimmed Mean of M-values (TMM) normalization, and differential expression analysis (DEA) was conducted using the R package limma (version 3.58.1). Linear models were fitted to the normalized data using the lmFit function from limma, with empirical Bayes methods applied via eBayes to stabilize variance estimates, crucial for datasets with small numbers of replicates. MiRNAs whose absolute log<sub>2</sub>-fold change was greater than 1 and whose false discovery rate (FDR)-adjusted p value was less than 0.05 were considered to be differentially expressed. To identify circulating

miRNA signatures capable of distinguishing between early-onset CAD patients, late-onset CAD patients, and healthy individuals, elastic net feature selection and random forest methods were used. Feature selection was conducted using the least absolute shrinkage and selection operator (LASSO) regression method. An elastic net model with an alpha value of 1 (corresponding to standard LASSO) was applied to the training data. The miRNA features were adjusted for clinical covariates (sex and age). Model discrimination was assessed by generating receiver operating characteristic (ROC) curves and calculating area under the curve (AUC) values using the pROC package.

## Results

*Patient baseline characteristics* are summarized in Table 1. Compared with the control group (58.62% and 10.34%, respectively), significantly more patients in the premature CAD group were male (85%) and smokers (40.0%). There were no significant differences in the prevalence of arterial hypertension or patient body mass index between these groups. Almost half (45%) of the early-onset CAD patients had a history of myocardial infarction (MI). The median LDL was significantly lower in the early and late CAD groups than in the control group (2.04 and 2.17 vs. 3.25 mmol/l, respectively), likely reflecting the higher prevalence of lipid-lowering medication use in the CAD groups. Patients in all study subgroups had at least one age- and sex-unrelated risk factor, with the early CAD group having 30.0% of patients with all four of these risk factors.

*Significantly differentially expressed miRNAs between patient groups.* A total of 287 different circulating miRNAs were analysed following sequencing and preprocessing. The miRNAs whose expression substantially changed within the study groups are shown in Fig. 1. Plot A shows a comparison between early-onset CAD and late-onset CAD, identifying 139 significantly differentially expressed (DE) miRNAs. Plot B reveals 123 significantly DE miRNAs between the late-onset CAD and control groups. Plot C shows only two significantly DE miRNAs (miR-454-5p and miR-130b-5p) between the early-onset CAD and control groups. In further analysis by three independent analytical approaches (LASSO, random forest, and DEA), a total of seven miRNAs—miR-10b-5p, miR-29c-3p, miR-142-5p, miR-320b, miR-451a, miR-486-3p and miR-625-3p—were identified as key biomarkers through differential expression analysis and feature selection. Among these, four miRNAs—miR-142-5p, miR-29c-3p, miR-451a, and miR-486-3p—were significantly downregulated in the late-onset CAD group compared with the control group. Additionally, all seven miRNAs demonstrated significant differential expression

**Table 1** Patient clinical and laboratory characteristics

| Variable                                                    | Control ( <i>n</i> =29) | Early CAD ( <i>n</i> =80) | Late CAD ( <i>n</i> =28) |
|-------------------------------------------------------------|-------------------------|---------------------------|--------------------------|
| Male, <i>n</i> (%)                                          | 17 (58.62%)             | 68 (85.0%)**              | 15 (53.57%)              |
| Smoking, <i>n</i> (%)                                       | 3 (10.34%)              | 32 (40.0%)**              | 4 (14.29%)               |
| Quitted smoking, <i>n</i> (%)                               | 1 (3.45%)               | 24 (30.0%)**              | 9 (32.14%)*              |
| Arterial Hypertension, <i>n</i> (%)                         | 20 (68.97%)             | 61 (76.25%)               | 23 (82.14%)              |
| History of percutaneous coronary intervention, <i>n</i> (%) | 0 (0.0%)                | 43 (53.75%)               | 18 (64.29%)*             |
| Previous myocardial infarction, <i>n</i> (%)                | 0 (0.0%)                | 36 (45.0%***)             | 16 (57.14%***)           |
| Age, years                                                  | 52.14±7.09              | 49.50 (IQR: 6.00)**       | 65.89±5.15***            |
| Body mass index, kg/m <sup>2</sup>                          | 33.28±4.74              | 29.93±4.60                | 29.39±4.90               |
| Total cholesterol, mmol/l                                   | 5.23±1.30               | 3.66 (IQR: 1.38)***       | 3.88 (IQR: 1.09)**       |
| LDL cholesterol, mmol/l                                     | 3.25±1.28               | 2.04 (IQR: 1.48)***       | 2.17 (IQR: 1.20)**       |
| HDL cholesterol, mmol/l                                     | 1.33 (IQR: 0.38)        | 1.07±0.30***              | 1.31 (IQR: 0.41)         |
| Triglycerides, mmol/l                                       | 1.36 (IQR: 1.15)        | 1.66 (IQR: 1.17)          | 1.58 (IQR: 1.21)         |
| Lipid lowering treatment, <i>n</i> (%)                      | 8 (32.0%)               | 70 (87.50%***)            | 26 (92.90%***)           |
| Risk factors <sup>a</sup> , <i>n</i> (%)                    |                         |                           |                          |
| 1                                                           | 8 (30.77%)              | 2 (2.50%)                 | 6 (21.42%)               |
| 2                                                           | 15 (57.69%)             | 27 (33.75%)               | 6 (21.42%)               |
| 3                                                           | 1 (3.84%)               | 27 (33.75%)               | 11 (39.29%)              |
| 4                                                           | 2 (7.69%)               | 24 (30.0%)                | 5 (17.86%)               |
| Creatinine, µmol/l                                          | 77.25±15.46             | 79.40±13.65               | 80.61±13.24              |
| Glucose, mmol/l                                             | 5.62±0.72               | 5.30 (IQR: 0.80)          | 5.25±0.73                |

<sup>a</sup>Presence of hypertension, smoking in anamnesis, obesity≥grade 1, hypercholesterolemia and/or lipid lowering drug use (age and sex were excluded to enhance comparability between the groups)

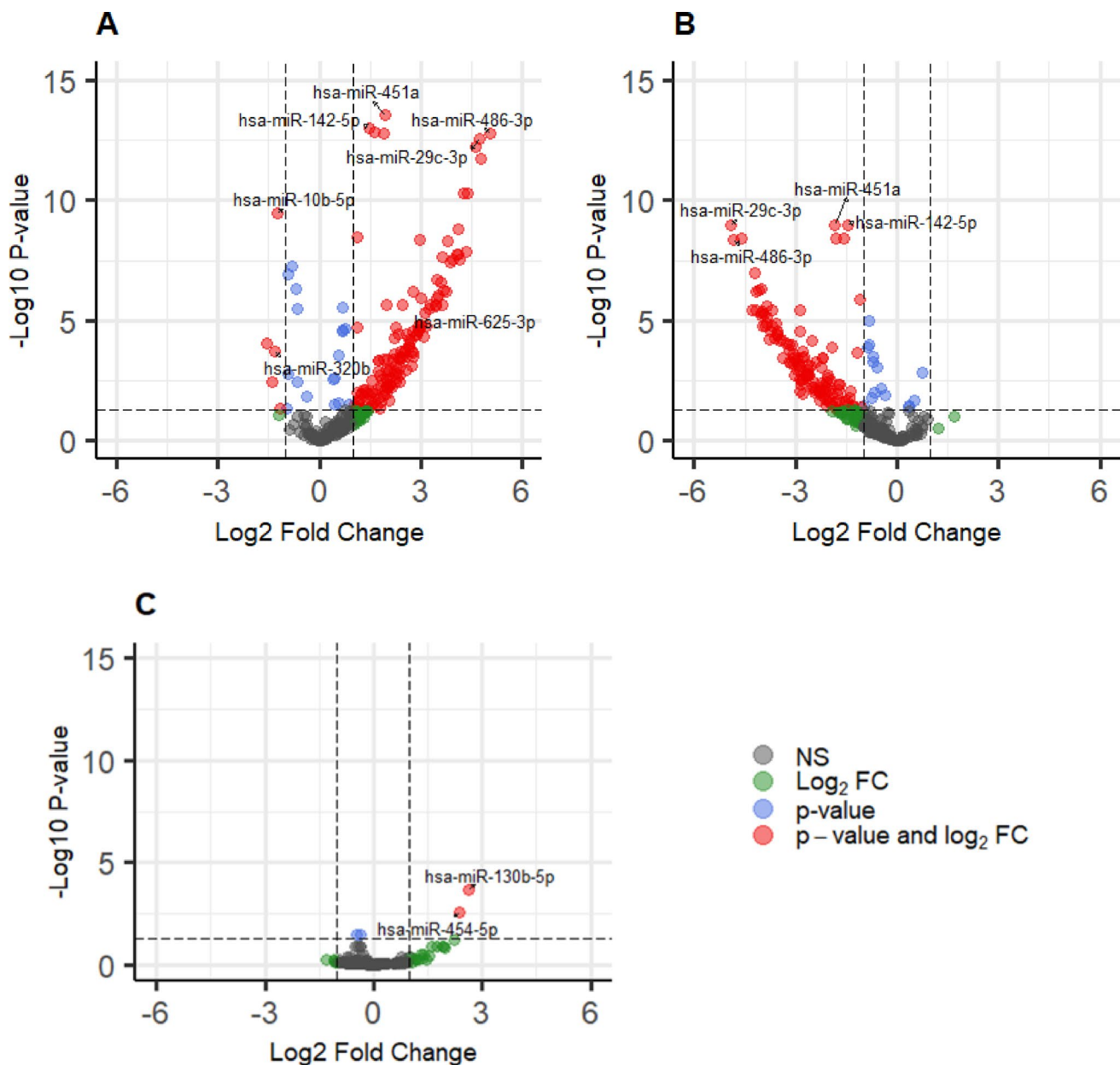
Statistical significance is denoted as follows: Represents statistically significant differences between early-onset and late-onset CAD cases vs. controls, whereas \*\*\*, \*\*, and \* indicate statistically significant differences ( $p<0.001$ ,  $p<0.01$ , and  $p<0.05$ , respectively) between Early-onset CAD and Late-onset CAD cases

between early-onset and late-onset CAD, indicating their potential biological relevance in distinguishing between different disease onset types (Fig. 2).

The seven miRNAs were considered the most robust and biologically relevant candidates and were used for the construction of stratified one-vs.-rest binomial logistic regression models adjusted for sex and age (i.e., a combined 7-miRNA model). Figure 3 shows the excellent predictive performance of the late-onset CAD vs. rest model, with an area under the ROC curve (AUC) of 0.9924, indicating near-perfect sensitivity and specificity (Table 2). Although the early-onset CAD vs. rest model also performed well, this model was less effective and less precise in distinguishing early-onset disease from late-onset disease. In contrast, the discriminative ability of the control vs. rest model was limited, with an AUC of 0.5304. Despite its high sensitivity (0.9386), its low specificity (0.2471) resulted in a modest accuracy of 0.7812 and a low kappa value of 0.1919, indicating weaker reliability. This suggests that while the model is highly sensitive for identifying healthy individuals, it also incorrectly classifies many CAD patients as healthy individuals.

## Discussion

Early-onset CAD is being increasingly recognized because of its ability to affect younger patients, leading to greater morbidity, a lower quality of life, and a substantial economic burden. Zeitouni et al. demonstrated that within a 10-year follow-up period, 52.9% of patients with premature CAD had at least one major adverse cardiovascular event (MACE), 18.6% had at least two recurrent MACEs and 7.9% had at least 3 recurrent MACEs, with death occurring in 20.9% of patients [3]. Furthermore, the recurrence of cardiovascular events is associated with increased health care utilization, such as hospital readmissions and emergency room (ER) visits. A study by Hsu et al. reported that the 1-year readmission rates following the index, first and second CAD recurrence events were 43.1%, 47.6% and 55.3%, respectively, and the proportions of patients who visited the ER were 46.4%, 51.9% and 57.8%, respectively [17]. The traditional risk factor burden is high among younger people [18, 19], with at least one risk factor present in 85–90% of patients with MI [19–21]. In addition, a tendency for the risk factor burden to increase has been observed in recent decades. Yandrapalli et al. reported that the prevalence of six modifiable risk factors—hypertension, dyslipidaemia, smoking, diabetes mellitus, obesity and drug abuse—increased in young adults with acute MI between 2005 and



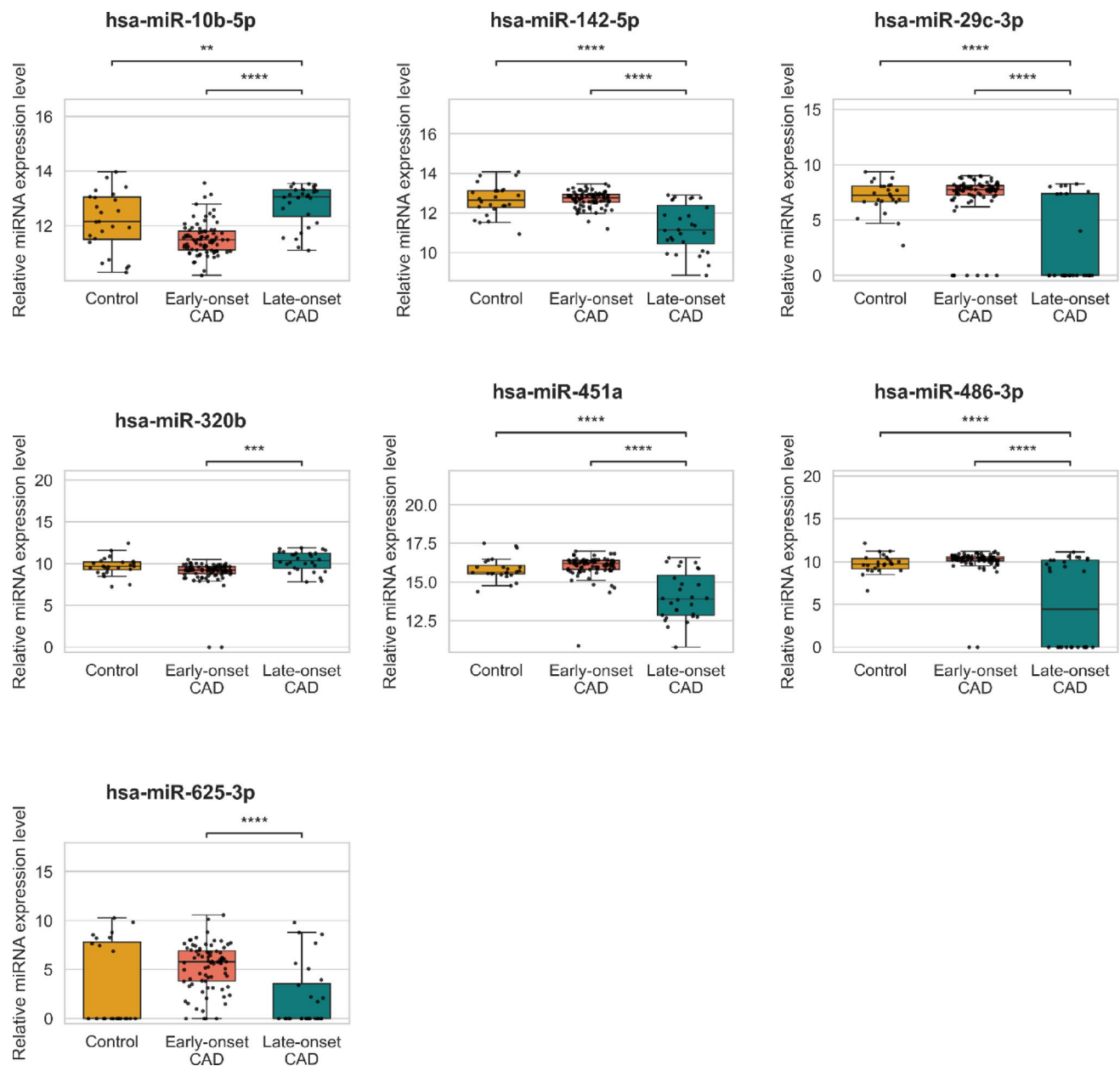
**Fig. 1** Differential Expression Analysis of miRNAs. Significantly altered miRNAs are marked in red. MiRNAs with the highest log<sub>2</sub> fold change are labeled in each plot. (A) Comparison

between early-onset CAD and late-onset CAD patients; (B) Comparison between late-onset CAD and the control group; (C) Comparison between early-onset CAD and the control group

2015, with a 98% relative increase in the prevalence of obesity in the 18- to 44-year-old patient group [19]. In line with the literature, the present study revealed a high prevalence of traditional risk factors across all groups, with the early CAD group exhibiting a great burden of risk factors, potentially contributing to the early development of CAD.

Several recent studies have indicated the potential role of circulating miRNA levels as valuable biomarkers for early CAD. Ying et al. demonstrated that the expression of four serum miRNAs (miR-196-5p, miR-3163-3p, miR-145-3p, and miR-190a-5p) was significantly decreased in

patients with very early-onset CAD and that they were able to differentiate these patients from controls [22]. Eren et al. analysed the expression of 13 miRNAs associated with endothelial cells, vascular smooth muscle cells, inflammation, and lipid metabolism among controls, early-onset CAD patients, and late-onset CAD patients. Their findings indicated that these 13 miRNAs exhibit distinct expression patterns in early CAD patients compared with both controls and late-onset CAD patients [23]. While earlier studies have focused primarily on the use of individual miRNAs as predictors of CAD, often yielding inconsistent results



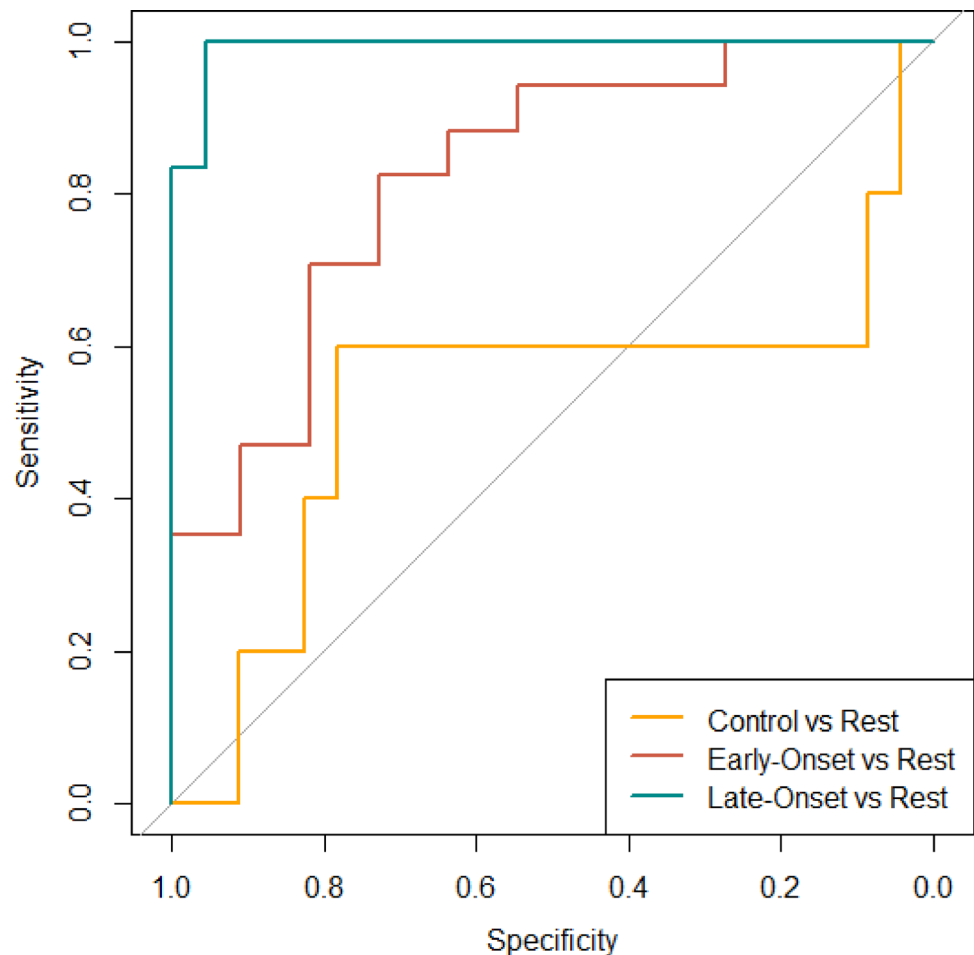
**Fig. 2** Expression of selected miRNAs across patient groups. TMM-normalized expression levels of selected miRNAs across control, early-onset CAD, and late-onset CAD groups with adjusted p-values

from differential expression analysis. Comparisons with adjusted  $p < 0.05$  are indicated by asterisks ( $p < 0.05$ : \*,  $p < 0.01$ : \*\*,  $p < 0.001$ : \*\*\*,  $p < 0.0001$ : \*\*\*\*)

across different studies, we aimed to assess the combined use of significantly expressed miRNAs within groups to improve the prediction of CAD development. In this study, we analysed the expression of 287 different circulating miRNAs and reported that their expression profiles significantly differed across our comparison groups. Our investigation revealed 139 miRNAs whose expression differed between early-onset and late-onset CAD patients and 123 miRNAs whose expression significantly differed between late-onset CAD patients and control patients. Among these, only two miRNAs—miR-130b-5p and miR-454-5p—were

notably upregulated in early-onset CAD patients compared with controls. The role of miR-130b-5p in CAD has been investigated in recent studies, highlighting its potential as a biomarker for this disease. Coban et al. reported that the expression of miR-130b-5p is negatively correlated with the complexity of CAD in female patients but is also positively correlated with plasma HDL levels and inversely correlated with fasting triglyceride levels [24]. Whereas miR-454-5p has been found to be upregulated in the platelets of CAD patients compared to healthy controls [25].

**Fig. 3** ROC curve analysis of one-vs-rest logistic regression models using 7 selected miRNAs adjusted for sex and age



**Table 2** Performance metrics of one-vs-rest logistic regression models using seven selected miRNAs adjusted for sex and age

| Class                | Accuracy | Sensitivity | Specificity | Kappa  | ROC-AUC |
|----------------------|----------|-------------|-------------|--------|---------|
| Control vs. Rest     | 0.7812   | 0.9386      | 0.2471      | 0.1919 | 0.5304  |
| Early-onset vs. Rest | 0.8106   | 0.7118      | 0.8897      | 0.6014 | 0.8235  |
| Late-onset vs. Rest  | 0.9854   | 1.000       | 0.9267      | 0.9508 | 0.9924  |

In our further analysis, seven differentially expressed miRNAs (miR-10b-5p, miR-29c-3p, miR-142-5p, miR-320b, miR-451a, miR-486-3p, and miR-625-3p) were identified between controls, early-onset CAD patients and late-onset CAD patients. Although the evidence is limited, the existing data suggest that these seven miRNAs may play a role in the development of CAD. MiR-10b-5p is associated with CAD through its role in lipid metabolism [26]. A study by Ye et al. suggested that miR-29c-3p may play a role in the formation of atherosclerotic plaques by regulating the proliferation and migration of vascular smooth muscle cells [27]. The downregulation of miR-142-3p has been linked to the inhibition of endothelial cell apoptosis and the suppression of atherosclerosis development [28], whereas the overexpression of miR-142-3p improved cardiac function and reduced inflammation in pig heart models with coronary microembolization [29]. Downregulation of miR-320b

is seen in patients with acute MI compared with controls [30, 31] and in patients with carotid plaques [32]. Compared with that in controls, the expression of miR-451a in CAD patients is upregulated, and along with miR-133a-3p, it serves as a strong biomarker for detecting CAD [33]. A study by Xu et al. demonstrated that miR-451a expression was significantly increased in acute MI patients compared with unstable angina patients and healthy individuals, suggesting its potential as a predictor for acute MI diagnosis [34]. Research has indicated that miR-486-3p could serve as a biomarker for STEMI. Compared with that in controls, the expression of miR-486-3p in patients with acute MI was upregulated [35], whereas Wei et al. demonstrated that even three months post-STEMI, the expression of miR-486-3p could reliably differentiate these patients from individuals with stable coronary artery disease [36]. Although research on the role of miR-625-3p in CAD is limited, there is

evidence suggesting its potential involvement in colorectal cancer [37].

To assess the diagnostic accuracy of these seven miRNAs in predicting CAD onset during early versus late stages, ROC curve analysis was performed. These findings indicated that the combination of these seven miRNAs had near-perfect sensitivity and specificity for distinguishing patients with late-onset CAD. However, their effectiveness and precision were reduced when early-onset disease was differentiated, and the model showed limited reliability in distinguishing controls from CAD patients. The potential of miRNA combination to improve CAD diagnostic precision was evaluated in a study by Zhang et al. In this study, they reported that plasma miR-29a-3p, miR-574-3p, and miR-574-5p were upregulated in CAD patients compared with controls. ROC curve analysis demonstrated that the combined use of these three miRNAs had greater discriminatory power than when each was evaluated individually [38].

This study has several limitations worth noting. The relatively small sample size in each group may have limited the final results. In addition, the early CAD group had a significantly smaller number of female patients with CAD. Therefore, the study data should be interpreted with caution, and further validation using large-scale studies is needed. The SYNTAX score was not calculated for patients with CAD, which could have provided additional insights into the complexity and extent of coronary artery disease among the participants. With respect to age, the study design stratified patients into early-onset CAD, late-onset CAD, and non-CAD controls with ages similar to those in the early-onset CAD group. To properly compare the late-onset group, additional individuals within the same age range as late-onset CAD are needed.

## Conclusion

The results of our study revealed differentially expressed miRNAs across study subgroups, providing novel insights into the potential role of circulating miRNA expression associated with early and late CAD onset. Our study identified a panel of seven miRNAs that effectively distinguished patients with late-onset CAD, with moderate accuracy in differentiating patients with early-onset CAD. These findings suggest that combined miRNA signatures hold promise for identifying early-onset CAD patients in the future. Further research in larger populations is needed to enhance the predictive capacity of these biomarkers for early disease detection and to develop more precise biomarkers for all stages of CAD.

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**Author contributions** E.K. was involved in study design, patient inclusion, blood sample preparation for miRNA expression analysis, data collection, manuscript writing. V.D. was involved in bioinformatics analysis, statistical analysis, participated manuscript writing. B.V. was involved in bioinformatics analysis, statistical analysis, manuscript editing. L.G. contributed in bioinformatics analysis, manuscript review. L.C. contributed in patient inclusion, data collection, manuscript editing. A. Ē did manuscript review. K.T. was involved in study design, patient inclusion, manuscript review, editing.

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**Data availability** No datasets were generated or analysed during the current study.

## Declarations

**Conflict of interest** The authors declare they do not have anything to disclose regarding conflict of interest with respect to this manuscript.

**Ethics approval** The study was performed in line with the principles of the Declaration of Helsinki. Approval was granted by Central Medical Ethics committee of Latvia (protocol No 2019-1). The study protocol was approved by Pauls Stradins Clinical university Institutional review board.

**Consent to participate** Informed consent was obtained from all individual participants included in the study.

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